Simple Pulsed Nuclear-Magnetic Resonance Spectrometer*

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A relatively simple, incoherent, variable frequency pulsed nuclear-magnetic resonance spectrometer is described. It uses mostly commercial units and can examine materials with T_2's greater than 50 μsec.

INTRODUCTION

QUITE a few articles describing pulsed nuclear-magnetic resonance (pnmr) spectrometers have been published,[1-6] all of which seem too complicated for a senior student laboratory. Since we wishec to use such a device in our senior student laboratory, we decided to design and build as simple and versatile a one as we could. The resulting pnmr spectrometer has been used by our senior students for the past year with good results. It operates well enough to also bc used in some of our own experimental work. This apparatus is described below. Besides its relative simplicity, its virtues are that its gating circuits are dc coupled, its frequency can bc varied easily over wide ranges, and it uses mostly commercial units. Rf power and quick recovery times have not been strived for in this apparatus. It can, however, be used to study liquids and solids with T_2's greater than 50 μsec.

I. APPARATUS

A block diagram of our pnmr apparatus is given in Fig. 1. Since the theory and experimental techniques of pnmr have been reviewed elsewhere,[7] we do not give the pnmr purpose of each of the blocks here.

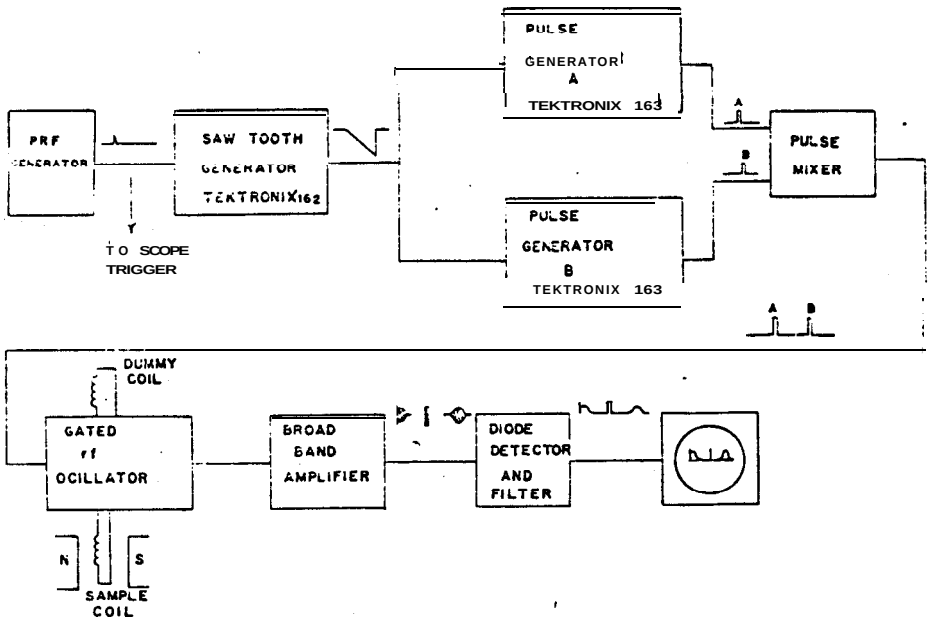


FIG. 1. Block diagram of pulsed nuclear magnetic resonance apparatus

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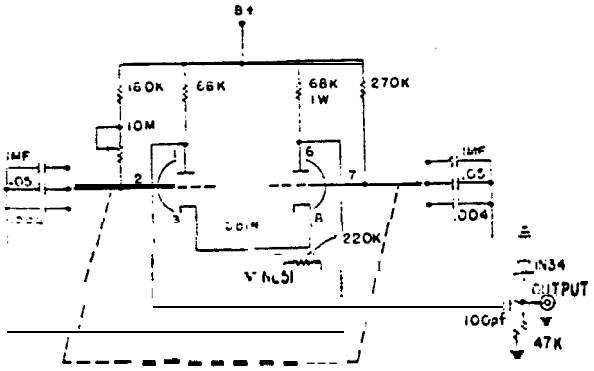


FIG. 2. Pulse-repetition frequency (prf) multivibrator. It covers prf range of 1 msec to 35 sec in three steps. B+ = 225V.

A. Pulse Programing Circuits

The pulse programing circuits are built around a **set of Tektronix**[8] pulse generators;

1 – 160 A power suppiy,
1 – 162 waveform generator,
2 – 163 pulse generators.

The 160 A power supply provides power for the prf gcncrator, pulse mixer, and gated rf oscillator as well as for the Tektronix components.

The prf generator is a free running multivibrator that generates a pulse to trigger the sawtooth generator. A complete pnmr experiment is carried out for each puise put out by the pri generator and thus irs repetition rate controls the number of pnmr experiments carried out per second. The Tektronix sawtooth generator has a built in prf generator that may also be used, but with the following severe limitation. In many experiments, one wishes to use a low repetition rate for the experiment while having a steeply descending sawtooth from the waveform generator so as to place pulses A and B close together. We found that this could not be done without using the auxiliary prf generator. Figure 2 is a schematic of the prf generator. The neon bulb flashes at the initiation of each pulse sequence.

The pulses from pulse generators A and B are mixed in the pulse mixer, whose schematic is given in Fig. 3. The 6BQ7 is biased well beyond cutoff and provides a – 1 V output signal to cut off the transistor in the gated rf oscillator during the absence of a gating pulse.

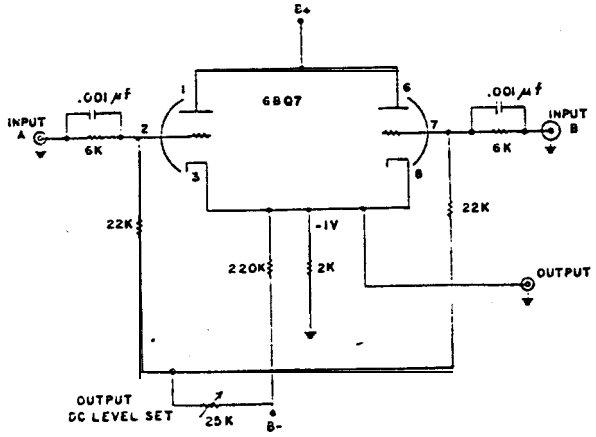


FIG. 3. Pulse mixer. B+ = 225 V, B– = – 170 V.

[8] Tektronix Corportation. P.O. Box 500, Beaverton, Ore.

B. Pulsed Radio-frequency Oscillator

The schematic for the pulsed radio-frcqucncy oscillator is shown in Fig. 4. It is simple in design,

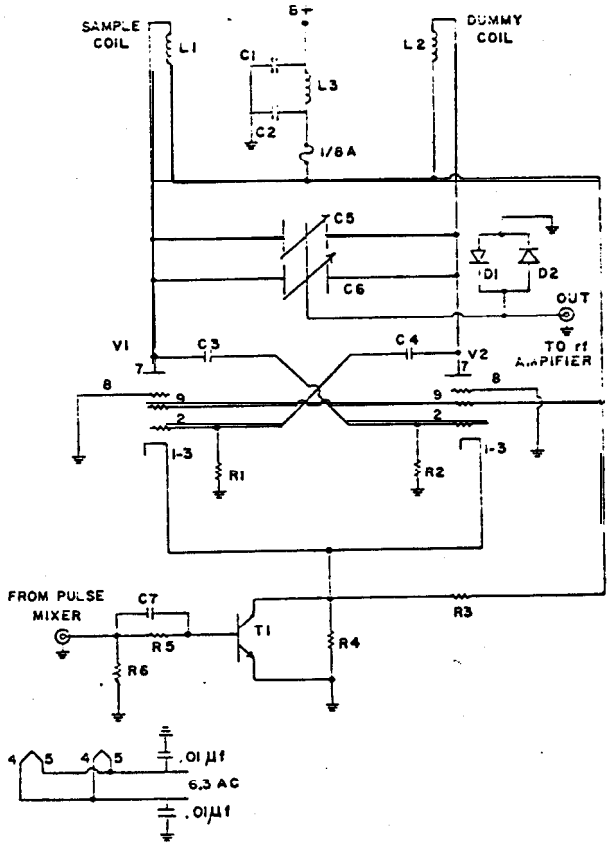


FIG. 4. Pulsed radio-frequency oscillator. $V_1 = V_2 = 6688$; $T_1 = 2N697$; $D_1 = D_2 = 1N1604$; $R_1 = R_2 = R_5 = 10$ k; $R_3 = 470$ k; $R_4 = 22$ k; $R_6 = 1$ k; $C_1 = C_2 = C_7 = 0.01$ μF; $C_3 = C_4 = j$ pF (mica); $C_5 = 6$ 8-to 51-pF butterfiy capacitor; $C_6 = 1.8$ to 5.1-pF differential capacitor; $L_3 = 100$ μH; L_1 and L_2 are the sample and dummy coils, respectively, whose values are chosen to give the proper resonant frequency; B+ = 200 to 350V.

but provides adequate power to examine liquids (... long T_2 solids. The oscillator two tubes in a symmetric configuration, each tube providing a 180° phase shift of the rf voltage around the feedback loop. The oscillator tubes are normally biased well beyond cutoff by the positive voltage on their cathoes and the circuit cannot oscillate. This bias is provided by the bleeder resistors R_3 and R_4 when the transistor T_1 is cut off. When an rf pulse is dcsircd, a gating pulse from the pulse mixer drives transistor T_1 into saturation, bringing the cathoes of V_1 and V_2 within a couple tenths of a volt of ground, thus turning the oscillator on. At the end of the gating pulse, transistor T_1 stops conducting and the cathode voltage returns to its original value. This value must bc well beyond the cutoff voltage of the tubes to make the oscillator stop oscillating.

The two identical coils L_1 and L_2, that resonate with capacitors C_5 and C_6 and determine the oscillator frequency, arc also the sample coil and dummy coil of the apparatus. The frequency of the oscillator (and thus of rhc whole apparatus) is easily varied by adjusting capacitor C_5 and coils L_1 and L_2. Our coils were 1 cm in diameter. For a 50-turn coil, we were able to cover the frequency range of 4.5 to 6.6 Mcps, and for a 3-turn coil we covered the range of 26 to 40 Mcps. Our construction techniques introduced a large amount of distributed capacity that would have to bc reduced if we were to go above 40 Mcps.

The cross-coupling capacitors C_3 and C_4 were found to have an optimum value of 5 pF. Larger values decreased the rf voltage across L_1 and L_2, and thus decreased the rotating magnetic field B_1. Smaller values for C_3 and C_4 tended to distort the rf pulse shape and to reduce the signal to noise ratio. The peak rf voltage across L_1 and L_2 under optimum conditions was 170 V, which corresponded to a 7 G rotating field at 4.7 Mcps and a 3 G rotating field at 30 Mcps. The rate of decay of oscillations of the oscillator were such that no signai from a sample could be seen for rhe 30 μsec following an 8 μsec pulse at 4.7 Mcps.

When the rf pulse is on, there is a direct current passing through coils L_1 and L_2, which produces a dc magnetic field B_2 perpendicular to the applied static magnetic field B_0. B_2 is of the same order of magnitude as B_1 and several orders of magnitude less than B_0. Its main effect is to change the direction of the applied static magnetic field by the angle $\theta \approx B_2/B_0$. This effect can be neglected.

The center plates of the butterfly capacitor C_5 and differential capacitor C_6 arc floating, and it is from this point that the output signal is taken. Since the voltages across L_1 and L_2 are 180° out of phase, under balanced conditions, there should be no transmitting of the rf pulse through the capacitors to the output. Due to lack of exact symmetry, one does find part of the rf pulse at the output. This is minimized by the balancing differential capacitor C_6 and the clipping diodes D_1 and D_2. To prevent rf pickup and and power supply noise reaching the output, components C_1, C_2, and L_3 filter the B + voltage supplied to the oscillator.

Attempts were made to improve the signal to noise ratio by matching the output impcdancc of the pickup circuit (made up of L_1, L_2, C_5, and C_6) to the 90-Ω impcdancc of the coaxial line and rf amplifier. We were not successful, which leads us to bclicvc that we are already approximately matched without the addition of any further components. A crude calculation of the output impedance is consistent with this result.

C. rf Amplifier and Detector

A model CT 50B rf amplifier made by RHG Electronics Laboratory, Inc.[9] was used in the apparatus. It is completely transistorized, has a gain of 60 dB and covers the range of 1 kcps to 50 Mcps. Its noise figure is 5 dB. Its good low-frequency response is obtained by the use of large coupling capacitors between stages. These capacitors charge up while the rf pulse is on and their slow discharge obscures the pnmr signal. This can be corrected by reducing all interstage coupling capacitors to 1000 pF.

The radio-frequency signal was detected using a 1N34 diode, and noise was filtered out by an rc network. To prevent overdriving the amplifier of the oscilloscope by the ri pulse, we used a 1N1604 silicon diode at the input to the oscilloscope to short large signals to ground.

Q RHG Eicctronics Laboratory, Inc. 94 Milbar Blvd., Farmingdale, N. Y.

D. Construction

Figure 5 is a Photograph of the apparatus. The prf generator and pulse mixer were constructed on a chassis that fit underneath the Tektronix pulse generator set. The pulsed rf oscillator was built into a small box so that it could be placed near the magnet producing the B_0 field. Due to the broad band-width of the rf amplifier, the pulsed rf oscillator had to be well shielded from a wide spectrum of rf noise. This was done by placing two aluminum chassis (each 7 in. X5 in. X3 in.) back to back (resulting in a box 7 in. X5 in. X6 in.), and using the common side as a mounting board. The tubes V_1 and V_2, and the floating variable capacitors C_5 and C_6 are then well shielded.

The coils L_1 and L_2 are mounted in small aluminum boxes (2 in. X3 in. $\times\frac{3}{4}$ in.) and connected to the oscillator box through dual coax cables. One needs to use dual coax cables for this connection because both ends of each coil are above ground.

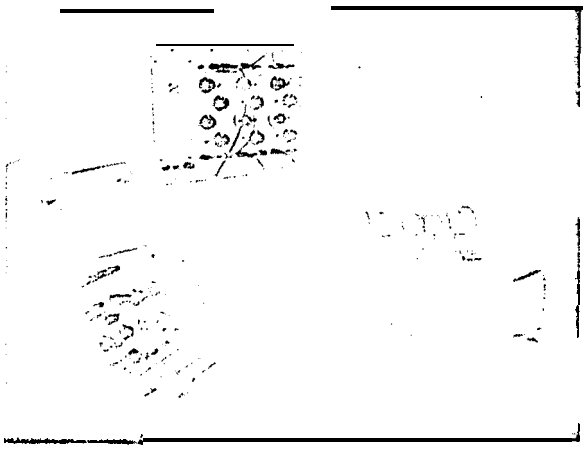


FIG. 5. PNMR spectrometer set up in student laboratory.

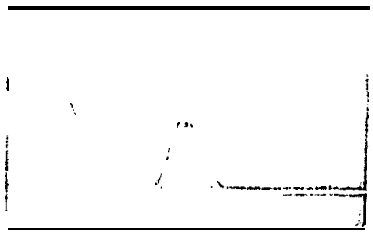


FIG. 6. Signal from a one cubic centimeter, doped water sample. Sweep speed -0.5 msec per cm : Scope gain -0.05 V per cm; B_0 = 7050 G. First pulse width $\approx 23\ \mu$sec; Second pulse width $\approx 43\ \mu$sec; Oscillator $B+ = 200$ V; L_1 and L_2 are one cm in diameter and have 3 turns.

II. OPERATION

For student laboratory use, a small permanent magnet (see Fig. 5) provided a B_0 field of about 1 kG for the pnmr apparatus. The unit yielded a signal whose signal to noise ratio was about 12 to 1 for a water sample filling the coil. When the apparatus was used with an electromagnet producing a field of 7 kG, the signal to noise ratio was much better than 20 to 1 for the same water sample. A typical signal for the latter case is shown in Fig. 6.

The only precaution to be taken in using the apparatus is that the $B-$ voltage be turned on before the $B+$ voltage so that that oscillator is cut off when the $B+$ voltage is turned on. Otherwise transistor T_1 and tubes V_1 and V_2 can be damaged.

ACKNOWLEDGMENT

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